

Augmenting the Historical Record: A Statistical Analysis of Global, Regional, and Domestic Climate Drivers of Sri Lankan rainfall



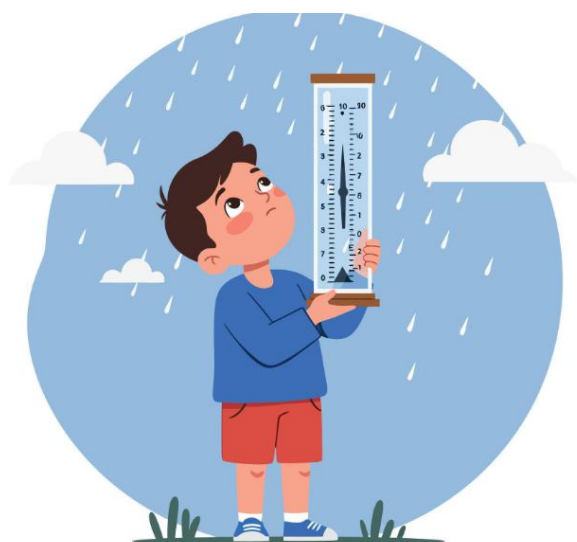
GROUP 2

1. s16948- Akindu Liyanage
2. s16974- Ayaz Ahamed
3. s16629- Dilmi Nawodya
4. s16806- Pasindu Jayasundara

Contents

1.	INTRODUCTION	3
1.1	The Sri Lankan Climate.....	3
1.2	Problem Definition: The Need for Multivariate Augmentation	3
1.3	Study Objectives and Scope.....	4
2.	DATA DESCRIPTION	5
2.1	Dataset Strategy	5
2.2	Data quality issues and preprocessing	5
2.3	Dataset Augmentation.....	5
2.4	Dataset Description.....	6
3.	EXPLORATORY DATA ANALYSIS	8
3.1	Univariate Analysis.....	8
3.1.1	Target variable (Precipitation).....	8
3.1.2	Predictors	8
3.2	Bivariate Analysis	10
4.	ADVANCED STATISTICAL ANALYSIS	12
4.1	Objective 1: Global-Scale Indicators:.....	12
4.1.1	Long-Term Trend and Stationarity Analysis for Rainfall	12
4.1.2	Niño 3.4 SST	13
4.1.3	Southern Oscillation Index	13
4.1.4	Sunspot Number.....	14
4.2	Objective 2: Regional-Scale Indicators:.....	14
4.2.1	DMI (Indian ocean dipole) - Influence of DMI on Sri Lankan Rainfall	14
4.2.2	All India Rainfall - Influence of All Indian Rainfall on Sri Lankan Rainfall	15
4.3	Objective 3: Domestic/Local Indicators:.....	16
4.3.1	U-Wind.....	16
4.4	Objective 4: Spatio-temporal analysis:	17
4.4.1	Seasonal Migration of Rainfall	17
4.4.2	Seasonal Spatial Distribution.....	17
4.4.3	Mechanisms of Variability: The Wind-Rain Correlation Flip	18
5.	RESULTS AND DISCUSSION	19

5.1. The Stability of the Historical Rainfall Record:	19
5.2. Global Drivers:	19
5.3. Regional Drivers: Indian Ocean Dipole (DMI) and All-India Monthly Rainfall	20
5.4. Local Dynamics: The Dominance of the Westerlies	20
5.5. Spatio-Temporal Dynamics: The Rain Shadow and The Blue Wall	20
6. CONCLUSION	21
7. REFERENCES	21



1. INTRODUCTION

1.1 The Sri Lankan Climate

Sri Lanka's hydro-climatology is a complex system that is governed by atmospheric circulations and seasonally varying monsoon systems. As a tropical island nation, the country's agricultural stability, hydropower generation and disaster management are directly linked to the variability of its rainfall. Sri Lanka generates much of its power through hydro-power plants (approx. 35%) and is strongly reliant on agriculture to drive its growth (agriculture makes up 7.2% of the GDP of Sri Lanka). Thus, the accurate understanding and quantification of the factors that drive its hydro-climatology is of extreme importance. However, the analysis of rainfall in isolation-as a simple univariate time series-limits the ability to gain an understanding of the factors & mechanisms that drive its variability. In order to move beyond such a surface level analysis and to gain a deeper understanding of the external factors that govern the Sri Lankan rainfall systems, it is necessary to bridge the gap between local rainfall metrics and a robust multivariate global analysis.

1.2 Problem Definition: The Need for Multivariate Augmentation

A core problem addressed in this study is the "climatological isolation" of historical rainfall records. While it is possible to obtain rainfall data in the island from the recent past, this data often exists without the context of the global, regional and local drivers that generated it. Furthermore, there is a distinct lack of publicly available historical rainfall data.

This report addresses the question: *How do global, regional, and local factors affect rainfall in Sri Lanka?* By augmenting historical local rainfall data spanning 30 years (1970-2000) taken from 5 stations as well as data spanning 10 years (1990-2000) taken from 10 stations; with key climate indices, this study aims to statistically quantify and understand the influence of drivers at three distinct spatial scales:

- **Global Scale:** Planetary phenomena such as the El Niño-Southern Oscillation (ENSO) and solar cycles.
- **Regional Scale:** Indian Ocean patterns including the Indian Ocean Dipole (DMI) and proximal influences such as the All-India Rainfall (AIR) index.
- **Local/Domestic Scale:** Domestic influences such as wind circulation in the island.

1.3 Study Objectives and Scope

This study focuses on addressing four main objectives:

- **Objective 1 (Global-Scale Indicators):** To analyze the relationship between Sri Lankan rainfall and planetary drivers, specifically Sea Surface Temperatures (Nino3.4 SST from the Hadley Centre Sea Ice and Sea Surface Temperature dataset), the Southern Oscillation Index (Climatic Research Unit, University of East Anglia) and Sunspot Numbers (WDC-SILSO, Royal Observatory of Belgium)
- **Objective 2 (Regional-Scale Indicators):** To examine the influence of regional atmospheric dynamics, represented by the Dipole Mode Index (DMI- taken from the National Oceanic and Atmospheric Administration derived from the HadISST1.1 Dataset.) and All-India Rainfall data (All-India and Macro-Regional dataset- Indian Institute of Tropical Meteorology).
- **Objective 3 (Domestic/Local Indicators):** To investigate the association between domestic rainfall patterns and local wind systems by analyzing 850 hPa u-winds (zonal wind component at approx. 1.5km from sea level – extracted from the NOAA-CIRES-DOE 20th Century Reanalysis V3 dataset) in the local region.
- **Objective 4 (Spatio-temporal analysis):** To investigate the seasonal migration and geographical distribution of rainfall across Sri Lanka.

2. DATA DESCRIPTION

2.1 Dataset Strategy

This analysis utilizes two primary augmented datasets in order to ensure statistical robustness. For global and regional drivers (Objectives 1 and 2), which require long-term observation to detect cycles, the study employs a dataset spanning 30 years (1970–1999) with augmented rainfall data taken from 5 dry zone weather stations. For domestic comparisons (Objective 3) and considering the possibility of including the Dry Zone and Wet Zone responses, the study utilizes a dataset spanning 10 years (1990–1999), which includes continuous records for 10 weather stations in both the wet and dry zones.

2.2 Data quality issues and preprocessing

The rainfall data utilized in these datasets were taken from several datasets issued by the Department of Meteorology, Colombo, Sri Lanka which spanned several time periods (some datasets contained rainfall data from weather stations for the entire period 1900-2000 while others contained data for various other subset ranges of this time period). These datasets were stored primarily in the .dat file type and each file contained the rainfall data from individual weather stations as a time series along with the latitude and longitude of the station as well as the elevation of the station. Missing values in the dataset were flagged with a “M” in the dataset, while days in which trace rainfall occurred were flagged with a “T”. The data was extracted using a python script which extracted the rainfall, date, station name as well as the respective latitude, longitude and elevation for each station along with the “M” and “T” flags for the flagged data. This data was then merged into a singular dataset which was then augmented by including the global, regional and local climate driver variables. The missing values (514 values) were imputed using K-Nearest Neighbors imputation.

2.3 Dataset Augmentation

A brief overview of the specific domain information that was used to augment this dataset will be provided here. In order to capture global scale climatic patterns, we decided to focus on the El Niño-Southern Oscillation (ENSO) which is a recurring, 3–7-year climate pattern in the tropical Pacific Ocean, characterized by shifting sea surface temperatures (warm El Niño, cold La Niña, or neutral) and atmospheric pressure. “ENSO is one of the most important climate phenomena on Earth due to its ability to change the global atmospheric circulation, which in turn, influences temperature and precipitation across the globe.” (What is El Niño-Southern Oscillation (ENSO)? *National Oceanic & Atmospheric Administration*). The Southern Oscillation Index, which is a standardized index based on the observed sea level pressure (SLP) differences between Tahiti and Darwin, Australia was used as a measure of the atmospheric pressure variations corresponding to ENSO fluctuations. Further in order to capture the long-term effects of solar cycles (normally 11-year cycles) on rainfall at a global scale, sunspot numbers were also used.

In order to capture regional drivers, the dataset includes a measure of the Indian Ocean Dipole (IOD) which is a climate phenomenon caused by irregular oscillation in sea surface temperatures in the Indian Ocean. This drives weather patterns in the region but is a particularly important regional rainfall driver for Sri Lankan weather patterns. “Indeed, it has been identified that during the positive IOD rainfall increases and the negative IOD decreases the rainfall of the country. For instance, the wet zone of Sri Lanka experiences a significant rainfall increase during the positive IOD, which causes flash floods, damage to the agricultural sector, infrastructure and property damage.” (Indian Ocean Dipole (IOD) - Relatively New Natural Phenomenon Identified in Indian Ocean which is Influencing Various Ways to Sri Lanka: *Institute of National Security Studies*). Furthermore, since the Sri Lankan monsoon system is a part of the larger South Asian monsoon system, a measure of rainfall in India was also included in the dataset.

Finally, in order to study local drivers, the dataset includes a measure of local winds in the form of u-wind at 850hPa (zonal wind components at approx. 1.5km from sea level) covering the Sri Lankan region.

2.4 Dataset Description

While the analysis utilizes 2 datasets spanning different time ranges, the variables in each are common between the two. The “Dry Zone stations” dataset contains daily rainfall data spanning 30 years (1970-2000) and contains a total of 52553 data records from 5 different dry zone stations: Anuradhapura, Mannar, Puttalam, Trincomalee & Vavuniya. This dataset was primarily used to look for long term patterns in rainfall. The “1990’s stations” dataset contains daily rainfall data spanning 10 years (1990-2000) and contains a total of 34283 data records from 10 different wet & dry zone stations: Anuradhapura, Batticaloa, Galle, Hambantota, Katugastota, Mannar, Puttalam, Ratnapura, Trincomalee & Vavuniya. This dataset was primarily used to study local weather pattern drivers.

The original variables are as follows:

- Station_Name – Name of the station.
- Latitude – Latitude of the station.
- Longitude – Longitude of the station.
- Elevation – Elevation of the station in meters from sea level.
- Date – Date of the observation
- Year, Month, Day – Decomposed date values used for time series lagging purposes and imputation.
- Precipitation – Amount of rainfall recorded in mm.
- Flag – “M” for originally missing & “T” for trace rainfall (recorded as 0 in precipitation column for the purposes of this study)

The variables used to augment the dataset are as follows:

- **Global Scale:**

- I. Nino 3.4_SST – Average monthly sea surface temperature in the Nino 3.4 region in °C. This was used to represent the sea surface temperature component of ENSO and captures the Nino 3.4 region which is most commonly used to study ENSO patterns. (Extracted from the Hadley Centre Sea Ice and Sea Surface Temperature dataset)
- II. Southern Oscillation Index – Average monthly normalized pressure difference between Tahiti and Darwin, Australia. This was used to represent the atmospheric pressure variation component of ENSO (Extracted from the Southern Oscillation Index (SOI) 3 decimal place dataset made available by the Climatic Research Unit, University of East Anglia)
- III. Sunspot_Number – Daily number of sunspots (Extracted from the WDC-SILSO dataset, Royal Observatory of Belgium)

- **Regional Scale:**

- I. DMI – Average Monthly SST gradient between the western equatorial Indian Ocean and the south-eastern equatorial Indian Ocean in °C. This is one of the most used indicators for the intensity of the Indian ocean dipole. (Extracted from Dipole Mode Index (DMI) Monthly Time-series dataset - National Oceanic and Atmospheric Administration derived from the HadISST1.1 Dataset)
- II. All_India_Monthly_Rainfall_mm – Average monthly all India rainfall in mm. (Extracted from the All-India and Macro-Regional dataset- Indian Institute of Tropical Meteorology)

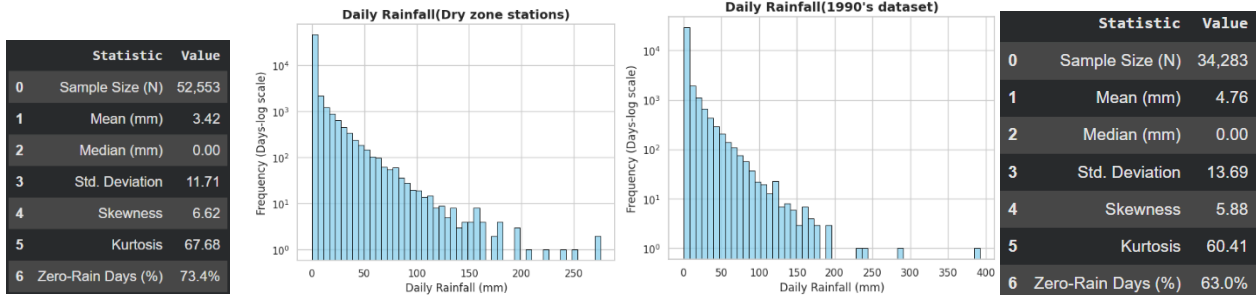
- **Local/Domestic Scale:**

- I. U_Wind_850_Mean – Average daily Zonal Wind (West-to-East wind component) at the 850 hPa pressure level (altitude of approx. 1.5km above sea level) taken from the grid point nearest to Sri Lanka (6°N, 80°E). (Extracted from the NOAA-CIRES-DOE 20th Century Reanalysis V3 dataset)

3. EXPLORATORY DATA ANALYSIS

3.1 Univariate Analysis

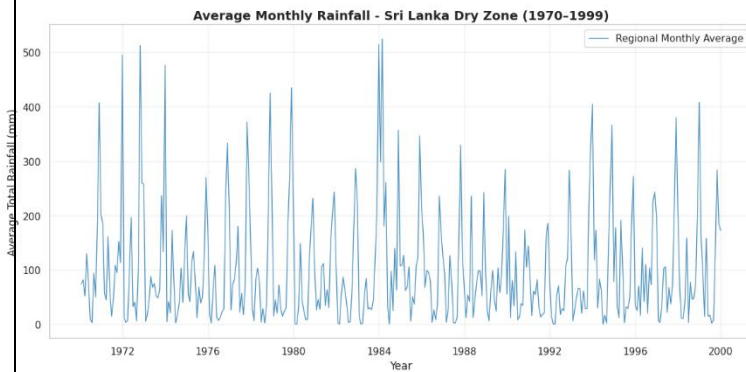
3.1.1 Target variable (Precipitation)



Histogram plots for the target variable in both datasets were obtained, with the y-axis in both histograms being log-scaled in order to better visualize the data in the tails (the histograms had very tall peaks). We can see that the target variable deviates from the normal distribution and instead follows a heavily right-skewed distribution in both datasets (Skewness is much greater than 0 in

both) and they also have large kurtosis values, indicating extreme leptokurtosis/heavy tails. This is due to a common effect in weather datasets known as ‘zero inflation’ which occur due the abundance of days with zero rain.

The line chart of average monthly rainfall does not appear to show any trend however it does seem to demonstrate seasonality.



3.1.2 Predictors

Since a majority of the study (Objectives 1,2 &4) were carried out using the “Dry zone stations” dataset, this section will mainly be focused on EDA carried out on this dataset.

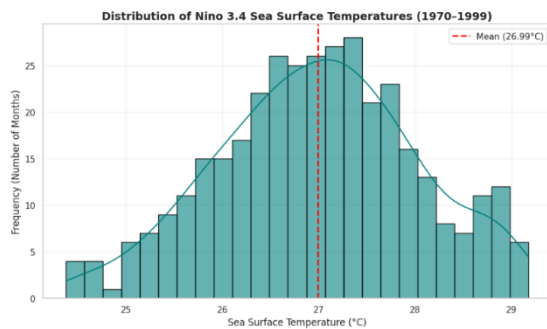
1.Rainfall by station

	Mean (mm)	Max (mm)	Std Dev	% Zeros
Station Name				
ANURADHAPURA (METEO)	3.34	149.1	10.58	71.0%
MANNAR	2.59	249.1	10.50	81.2%
PUTTALAM	3.22	275.7	11.43	72.7%
TRINCOMALEE	4.12	273.4	13.61	72.3%
VAVUNIYA	3.77	198.0	11.97	70.7%

Here we can see how rainfall is distributed across the 5 dry zone stations, as well as how “Dry” these areas really are. We can see, by the fact that the percentage of zero rain days is high for all of the stations, that these stations do in fact receive relatively low amounts of rainfall throughout the year. Trincomalee has received the highest

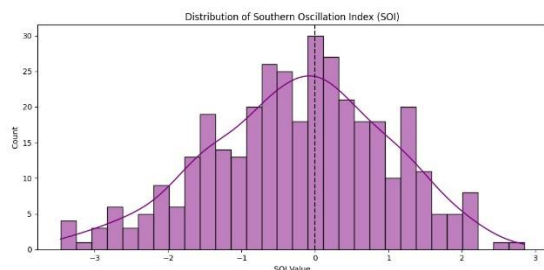
average rainfall across the 30-year period at 4.12 mm, while Mannar has received the least at 2.59 mm. It can also be seen that Mannar has been the “Driest” station throughout the period recording 81.2% zero rain days across the 30 years.

2.Nino 3.4 SST



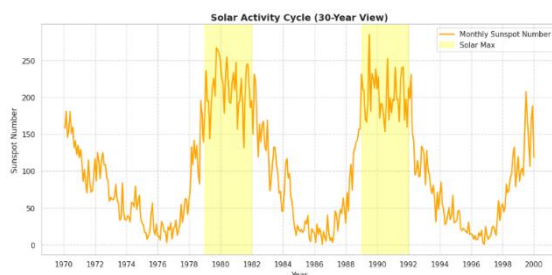
The distribution of the Niño 3.4 sea surface temperatures seems to approximately follow a normal curve with a slightly heavier right tail and a mean of 26.99°C. A departure from normality in these sea surface temperatures generally indicates the presence of stronger El Niño or La Niña phases. This will be further tested in the advanced analysis section of this report.

3. Southern Oscillation Index



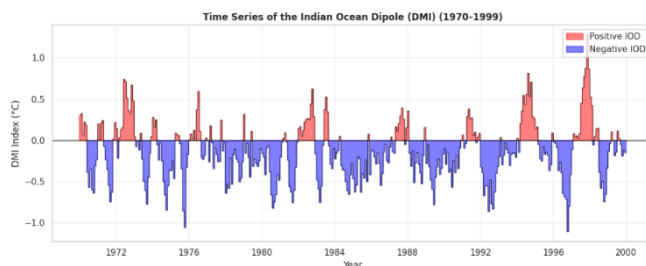
The distribution of the southern oscillation index appears to follow an approximately normal distribution centered at zero. Slight deviations can be seen in the mid sections of the tails, however as the data is normalized, we can assume that it roughly adheres to the normal distribution.

4. Sunspot number



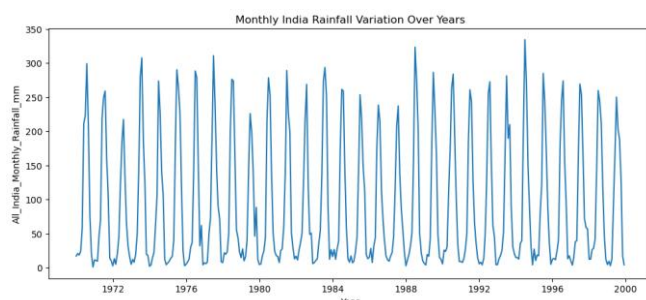
The line chart for the monthly sunspot number variable clearly emphasizes the solar cycle, with peaks at approximately every 11 years and clear troughs in between. While the cyclic pattern is evident, there is no apparent trend in the graph.

5. DMI (Indian ocean dipole)



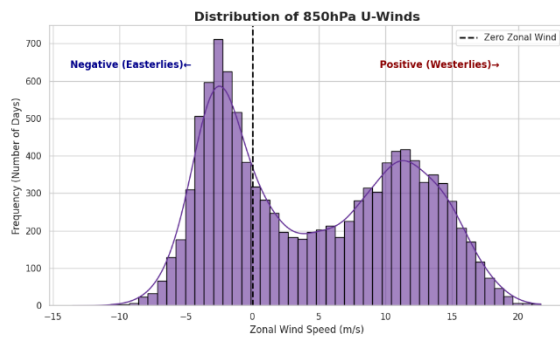
The line chart for DMI shows a clear oscillation in the Indian ocean dipole, alternating between positive IOD values and negative IOD values. This alternation is one of the key drivers for large amounts of rainfall as well as droughts in the region.

6.All India Rainfall



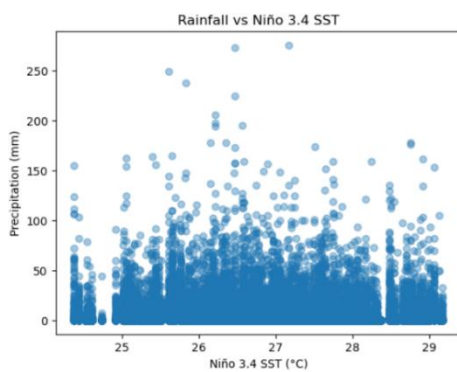
The seasonal nature of weather patterns in India is apparent in this line chart. Rainfall consistently peaks towards the middle of each year before decreasing towards the end of the year. No apparent cyclic patterns or trends can be seen in the data.

7.U-Wind

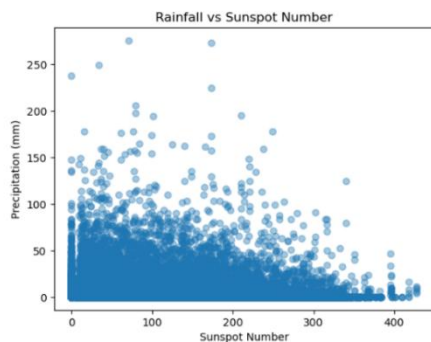


The distribution of u-winds shows a clearly bimodal distribution with separate peaks in both the negative and positive regions. This occurs due to the presence of both westerly winds (West to East blowing winds - given by positive u-winds) as well as the presence of easterly winds (East to West blowing winds given by negative u-winds)

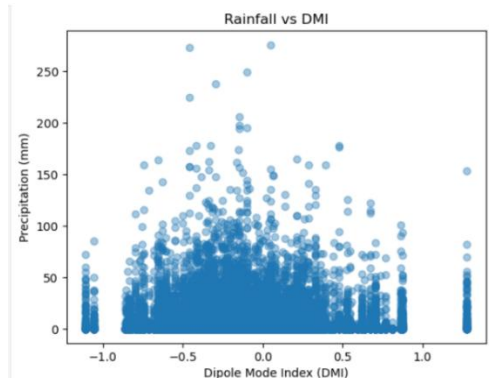
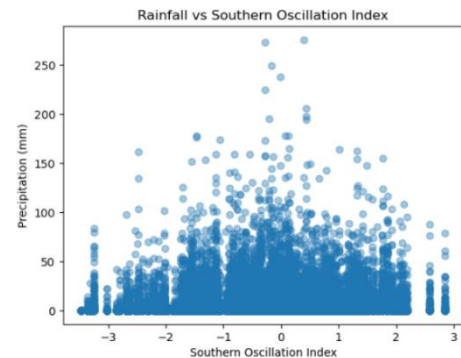
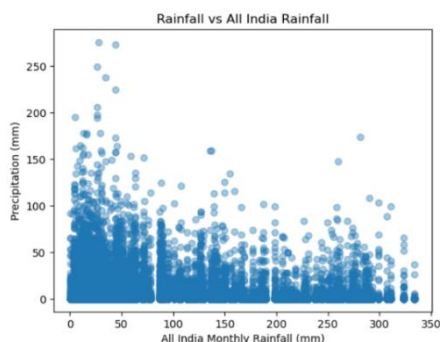
3.2 Bivariate Analysis



suggesting that ENSO sea surface temperature alone does not significantly regulate daily rainfall in Sri Lanka.

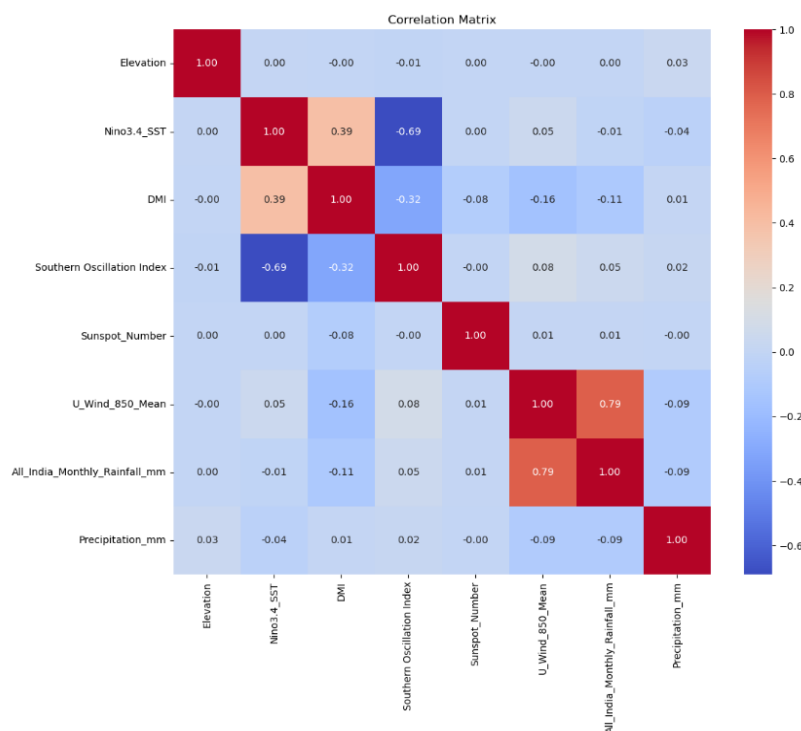
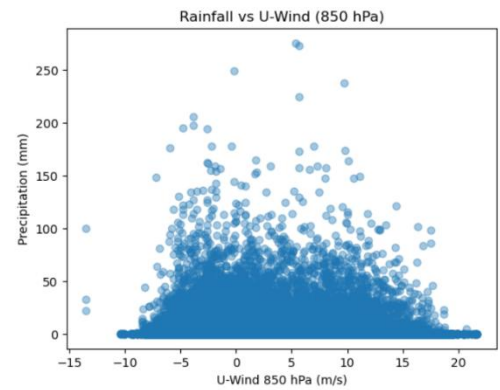


In the same way, the plot of rainfall against the Southern Oscillation Index displays significant scatter without a distinct trend, indicating that atmospheric pressure changes associated with ENSO do not solely account for rainfall variability. The examination of rainfall in relation to sunspot counts reveals no discernible trend, suggesting that solar activity exerts a minimal short-term effect on precipitation. Similarly, the plot of rainfall against the Dipole Mode Index seems to indicate slightly increased rainfall during negative DMI values, yet significant variability persists, implying a moderate regional effect of the Indian Ocean Dipole.



The plot comparing precipitation in Sri Lanka to All-India rainfall also does not indicate a clear pattern especially where one would expect that

increased rainfall in India would typically align with greater rainfall in Sri Lanka. The plot of rainfall against U-wind (850 hPa) indicates that increased rainfall predominantly takes place during westerly wind conditions, highlighting the significant influence of local wind circulation on rainfall patterns. In general, the bivariate analysis suggests that Sri Lankan rainfall is affected more significantly by local and regional elements rather than by specific global factors, and that variability in rainfall may arise predominantly from the joint impact of various climate influences.



The correlation heatmap indicates that precipitation has weak links with global, regional and local climate drivers – suggesting that these factors by themselves do not significantly affect daily rainfall in Sri Lanka – at least when non-lagged & non-seasonal values of these variables are considered.

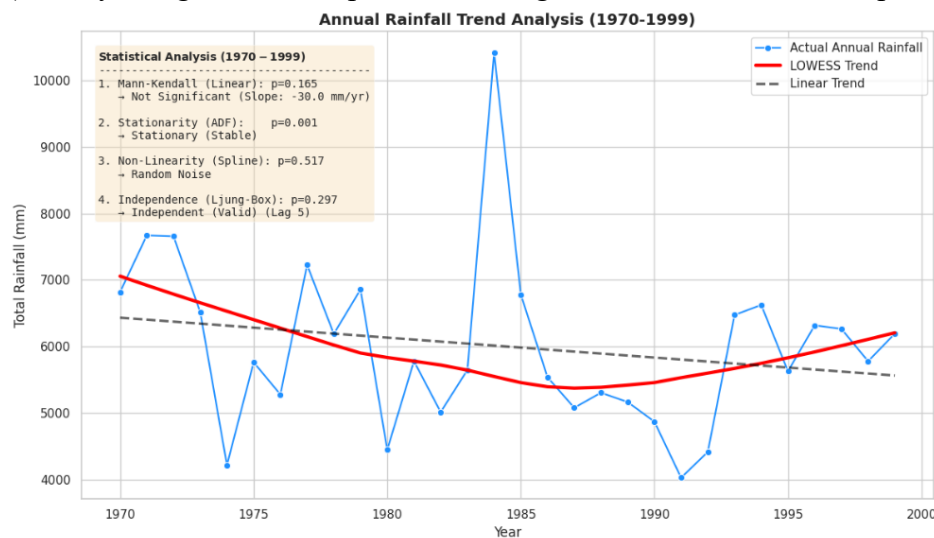
This suggests that deeper attention needs to be paid to the analysis of rainfall with lagged variable values as well as close attention should be paid to how these variables vary by season.

4. ADVANCED STATISTICAL ANALYSIS

4.1 Objective 1: Global-Scale Indicators

4.1.1 Long-Term Trend and Stationarity Analysis for Rainfall

As an initial step of the analysis, a line chart for annual rainfall was drawn and several tests were carried out to test whether the series is stationary and whether a trend is present in the data. The daily data was aggregated into annual totals, and a line chart was plotted. To check whether a linear or non-linear trend or pattern may be present in the data a linear as well as a LOWESS trendline (Locally Weighted Scatterplot Smoothing- creates a flexible, non-parametric trendline by fitting

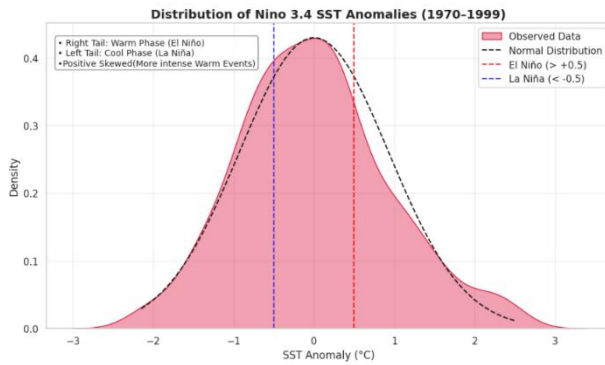


multiple linear regressions to localized subsets of data) were fitted and included in the plot.

A **Mann – Kendall test**- which is a non-parametric statistical method used to detect monotonic upward or downward trends in time series data was used to check for a

statistically significant linear monotonic trend as the linear trendline seemed to gradually decline. The Mann-Kendall test (which tests **H_0 : There is no monotonic trend in the data Vs. H_1 : A monotonic trend exists**) resulted in a **p-value of 0.165** which results in a **failure to reject the null hypothesis** and thus no statistically significant monotonic trend is present. While the Mann-Kendall test is not affected by the non-normality of our data (It does not require normality), it does require independence of data points, which is often an issue in time series data. In order to preserve the robustness of the test and to validate our conclusion, a **Ljung-Box test** (tests **H_0 : The data is independently distributed Vs. H_1 : The data exhibits serial correlation**) was applied for up to 5 lags (accounting for the 3–7-year ENSO cycle). This resulted in a **p-value of 0.2975** which once again resulted in a **failure to reject the null hypothesis**. This result ensures the robustness of our Mann-Kendall test conclusion. Furthermore, since the LOWESS trendline seemed to show a non-linear pattern, in order to test for such a complex, non-linear pattern, a spline regression model was fitted. However, the model **p-value was 0.517** which indicates that there is no significant non-linear curvature in the annual totals for rainfall. Finally, an **ADF test** was used to test the stationarity of the data and a **p-value of 0.0014**, leads to a **rejection of the null hypothesis** indicating that the series is stationary.

4.1.2 Niño 3.4 SST



As previously mentioned, a deviation from normality for the sea surface temperature generally indicates the presence of stronger El Niño or La Niña phases. Furthermore, by plotting the SST anomalies (SST - SST long term avg.) tells us how much hotter or colder a certain month is from the norm. A SST anomaly greater than +0.5 indicates a ‘hotter’ El Niño phase while one lower than -0.5 indicates a ‘colder’ La Niña phase.

Departures from normality in the form of skewness indicates a push toward one of these phases. A Shapiro-Wilk test confirmed the deviation from normality and a positive skew confirms that extreme El Niño events occur more frequently than would be expected in a purely random normal distribution. A Mann-Whitney U test was also conducted (Classifying SST anomaly as neutral if $-0.5 \leq \text{sst anomaly} \leq 0.5$ and as El Niño and La Niña otherwise based on the relevant criteria) to check whether either phase affected rainfall in a statistically significant manner. However, both of these resulted in showing that the phases were not statistically different from the neutral phase.

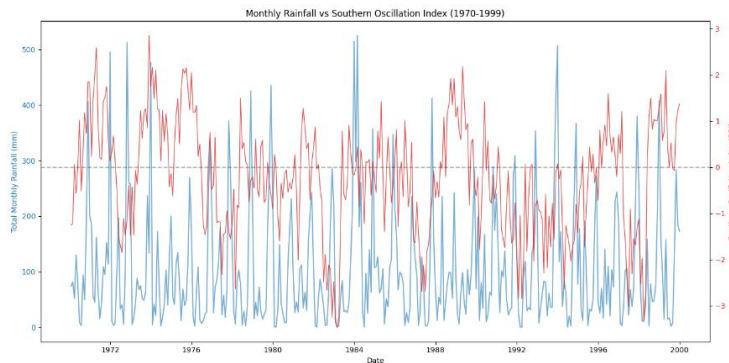
STATISTICAL DIAGNOSTICS

```
Skewness: 0.3373 (Positive: Right/El Nino skew)
Kurtosis: 0.0056
Shapiro-Wilk Test: p-value = 3.8011e-03
>> RESULT: The distribution is STATISTICALLY DIFFERENT from Normal.
```

Mann-Whitney U

```
El Niño vs Neutral: p-value = 0.3139 (Not Significant)
La Niña vs Neutral: p-value = 0.3206 (Not Significant)
```

4.1.3 Southern Oscillation Index



The analysis of the Southern Oscillation Index (SOI) reveals that it is not a strong or immediate driver of rainfall in Sri Lanka. Statistically, the **Pearson Correlation ($r = 0.038$, $p = 0.48$)** and the **Independent T-Test ($p = 0.43$)** both failed to reach significance, indicating that whether the SOI is in a positive (La Niña) or negative (El Niño) phase, the

immediate national-level rainfall remains largely unaffected. This lack of relationship is visually supported by the line chart, which shows a noisy pattern without consistent synchronization between SOI fluctuations and rainfall peaks. Furthermore, the **Lag Analysis** across a 12-month window showed no statistically significant results, confirming that atmospheric pressure fluctuations measured by the

```
2. T-Test (Positive SOI vs Negative SOI):
Mean Rain (Positive SOI): 101.93 mm
Mean Rain (Negative SOI): 106.79 mm
T-Statistic: -0.426
P-Value: 0.671
```

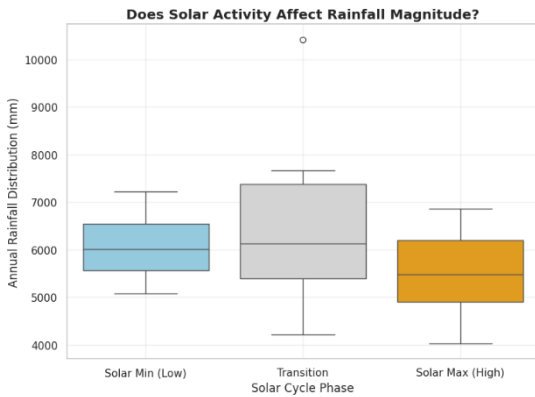
SOI are a weak and unreliable predictor for rainfall patterns in this context.

```
1. Pearson Correlation (SOI vs Rainfall):
Correlation: 0.0644
P-Value: 0.2226
```

3. Lagged Correlation Analysis (SOI predicting Rainfall):

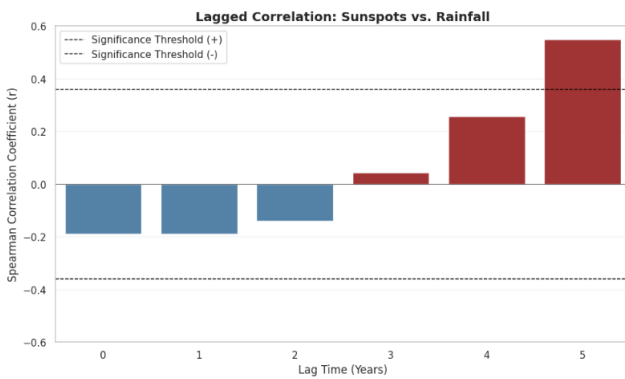
Lag (Months)	Correlation	P-Value
0	0.0644	0.2226
1	-0.0083	0.8749
2	0.0034	0.9490
3	0.0170	0.7491
4	-0.0143	0.7875
5	-0.0182	0.7323
6	-0.0380	0.4764
7	-0.0321	0.5480
8	-0.0678	0.2044
9	-0.0253	0.6364
10	-0.0623	0.2450
11	-0.0763	0.1548
12	-0.0632	0.2399

4.1.4 Sunspot Number



0.3433 & 0.6215 for the Shapiro-Wilk Test for the groups establishing normality in the groups and a Levene's test provided a p-value of 0.1037 showing that the variances across the groups are roughly equal. The Anova test

```
=====
ONE-WAY ANOVA REPORT: SOLAR PHASES (Max vs Min vs Trans)
=====
1. Overall ANOVA Test
Hypothesis H0: Mean rainfall is EQUAL across all 3 phases.
F-Statistic: 1.2029 | P-Value: 0.3159
-> Result: Fail to Reject H0 (No Significant Difference)
```



yielded a p-value of 0.3159 leading to a failure to reject the null hypothesis (Mean rainfall is equal across all 3 phases). However, calculating the correlation between rainfall and sunspots shows that annual rainfall is correlated with annual sunspots with a lag of 5 (spearman rank p-value:0.0045). This suggests that rainfall in Sri Lanka is influenced by solar activity in the past.

4.2 Objective 2: Regional-Scale Indicators

4.2.1 DMI (Indian ocean dipole) - Influence of DMI on Sri Lankan Rainfall

Hypotheses:

(H₀): (Null hypothesis): Rainfall (Negative DMI) ≤ Rainfall (Positive DMI)

(H₁): (Alternative hypothesis): Rainfall (Negative DMI) > Rainfall (Positive DMI)

```
--- Dry Zone Stations ---
Mann-Whitney U statistic: 302539110.0
p-value (one-sided, Negative > Positive): 0.006603927033936648
Sri Lankan rainfall is significantly higher during Negative DMI months in Dry Zone.
```

```
--- Wet Zone Stations ---
Mann-Whitney U statistic: 39057025.0
p-value (one-sided, Negative > Positive): 0.014901470668943689
Sri Lankan rainfall is significantly higher during Negative DMI months in Wet Zone.
```

To examine the influence of the Indian Ocean Dipole on Sri Lankan rainfall, a one-sided Mann-Whitney U test was conducted to test whether negative DMI (IOD) conditions are associated with

increased rainfall in Sri Lanka (as in reality). Rainfall during negative DMI months was compared with rainfall during positive DMI months separately for dry zone and wet zone stations.

For the dry zone stations, the test produced a p-value of 0.0066, indicating a statistically significant increase in rainfall during negative DMI months. Similarly, for the wet zone stations, the p-value was 0.0149, also below the 5% significance level, confirming that rainfall is significantly higher under negative DMI conditions.

These results provide strong evidence of a positive association between negative Indian Ocean Dipole phases and enhanced rainfall across Sri Lanka, affecting both dry and wet climatic zones.

4.2.2 All India Rainfall - Influence of All Indian Rainfall on Sri Lankan Rainfall

Test for dry stations Mann-Whitney U statistic: 133657360.0 p-value: 1.6201588421550795e-167 Sri Lanka rainfall differs significantly between high and low India rainfall months.
--

Test for wet stations Mann-Whitney U statistic: 20742287.0 p-value: 1.3224153922208675e-50 Sri Lanka rainfall differs significantly between high and low India rainfall months.
--

The purpose of this analysis was to examine whether high rainfall in India corresponds to increased rainfall in Sri Lanka, using data from both dry and wet zone stations. A Mann Whitney U test was applied to compare Sri Lankan rainfall during high versus low India rainfall months. We categorized India's monthly rainfall into High, Medium, and Low using the 67th and 33rd percentiles. Only the High and Low categories were used for testing.

(H₀): Sri Lankan rainfall does not differ between high and low India rainfall months.

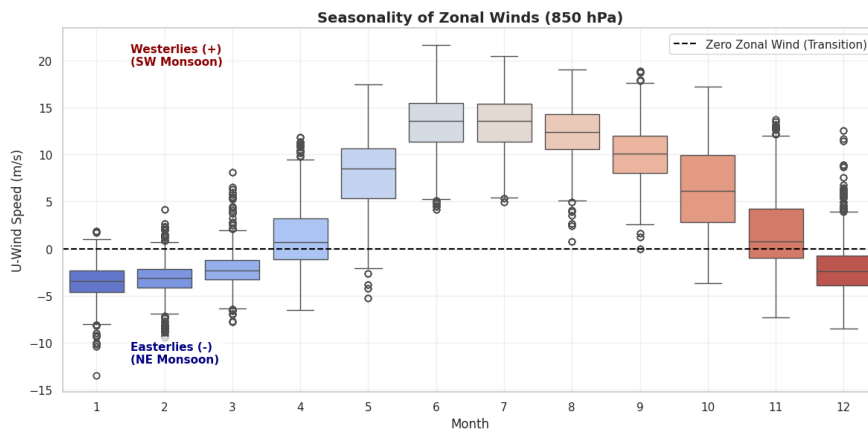
(H₁): Sri Lankan rainfall is higher during high India rainfall months.

For both dry zone stations and wet zone stations the test gave p-values less than 0.05, indicating a significant increase in Sri Lankan rainfall during high India rainfall months. These results suggest a strong positive association between India's monsoonal rainfall and Sri Lanka's precipitation. The southwest monsoon, which brings heavy rainfall to India, appears to influence both wet and dry zones of Sri Lanka, likely through regional atmospheric circulation and moisture transport across the Bay of Bengal and Indian Ocean.

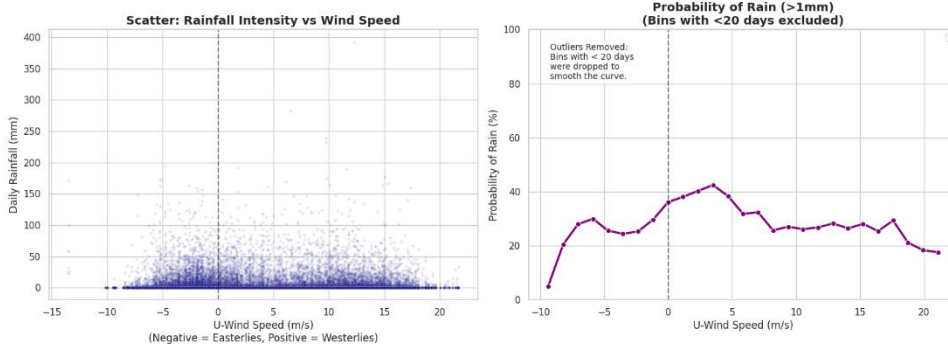
In conclusion, months of high Indian rainfall are associated with higher rainfall in Sri Lanka, confirming that the two rainfall patterns are interconnected, consistent with known monsoonal dynamics in the region.

4.3 Objective 3: Domestic/Local Indicators

4.3.1 U-Wind



The above figure shows the gradual shift from Westerly winds which are strong drivers of the South-Western monsoon (typically May to September) to Easterly winds which drive the North-Eastern monsoon period (typically December to February). The inter monsoons (October to November & March to April) are typically driven by a mix of both easterly and westerly winds as can be seen in the diagram (these periods are typically when the boxes cross the threshold.) The distribution of data here clearly reflects what is to be expected when compared to real world scenarios.



In order to construct the above plot, wind speed was separated into 30 bins and rainfall was classified as “rained” if rainfall > 1mm and “not rained” otherwise, within each of the bins. Probability of rain was then calculated using this classification and plotted with u-wind speeds & bins with less than 20 days were dropped in order to smooth the curve. The resulting plot shows 2 major spikes in probability of rainfall each in the Easterly and Westerly regions of the plot, suggesting that these may coincide with wind speeds that occur more commonly during monsoon periods. However, the probability of rain does not appear to significantly increase with extreme wind speeds and instead seems to be more strongly affected by moderate Westerly and Easterly wind speeds.

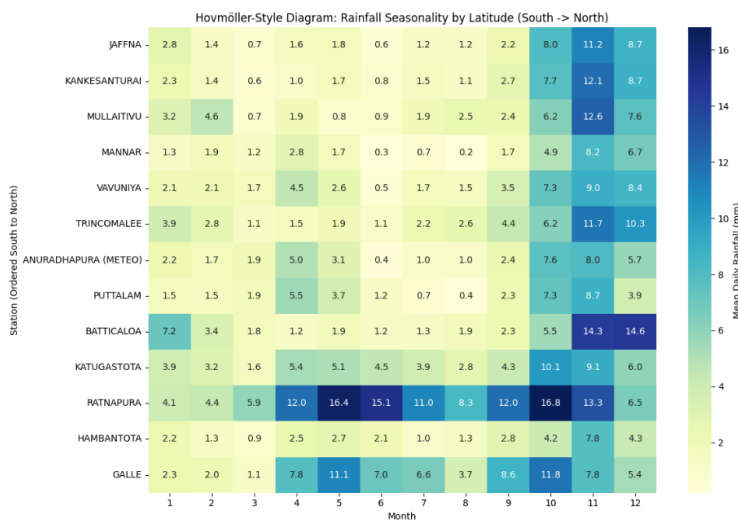
```
=====
STATISTICAL TEST: WESTERLIES VS EASTERLIES (Mann-Whitney U)
=====
HYPOTHESES:
H0 (Null):      Rainfall distributions are identical.
                (Median Rain_West <= Median Rain_East)
H1 (Alternative): Westerlies bring significantly MORE rainfall than Easterlies.
                (Median Rain_West > Median Rain_East)
=====
Mean Daily Rain (Westerlies): 4.98 mm
Mean Daily Rain (Easterlies): 4.41 mm
Mann-Whitney U Test p-value: 6.0757e-14
=====
>> RESULT: Reject H0.
Conclusion: Westerlies bring STATISTICALLY MORE RAIN than Easterlies.
=====
```

Easterly winds.

A Mann-Whitney U test was also carried out to test if Westerly winds bring statistically more rain than the Easterly winds (As is expected in real world situations) This gives a p-value much smaller than 0.05 resulting in the rejection of the null hypothesis in favor of the alternative hypothesis, indicating that Westerly winds do in fact bring more rain than

4.4 Objective 4: Spatio-temporal analysis

4.4.1 Seasonal Migration of Rainfall

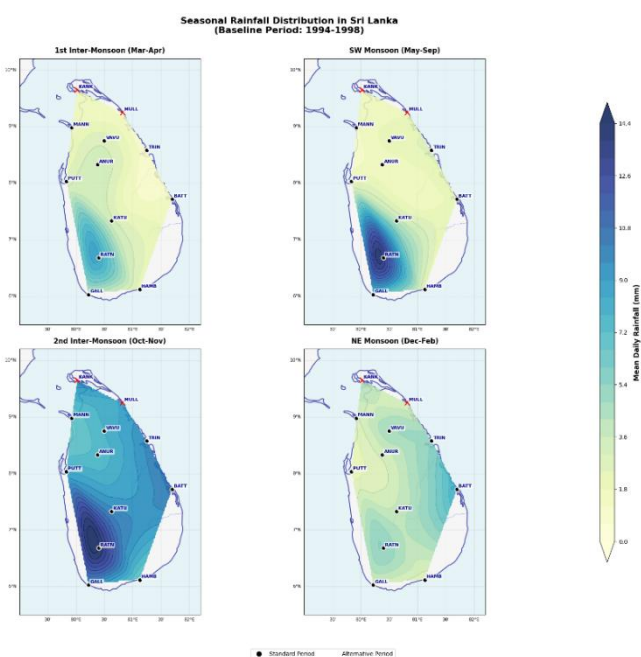


This Hovmöller-style diagram illustrates the **bimodal rainfall seasonality** of Sri Lanka, highlighting the geographical and temporal difference between the Southwest and Northeast Monsoons. The data reveals a clear "rain shadow" effect from May to August, where southern "Wet Zone" stations like Ratnapura and Galle experience intense precipitation—peaking at 16.4 mm daily—while northern stations like Mannar and Jaffna remain nearly parched, often dropping below 1.0 mm. This shifts dramatically during the Second Inter-monsoon and the beginning of

the Northeast Monsoon (October–December), creating a nationwide deluge of rainfall. During this period, the entire island receives significant moisture, including the arid North and the Eastern coast. Ultimately, the chart underscores that while the south enjoys more consistent year-round rain, the North and East are almost entirely dependent on the year-end monsoonal surge for their annual water budget. A Friedman test confirmed significant seasonality ($\chi^2 = 93.9395$, $p < 0.001$) and significant spatial variation ($\chi^2 = 52.9876$, $p < 0.001$), validating the visual evidence of monsoonal bimodality.

4.4.2 Seasonal Spatial Distribution

This spatial analysis of Sri Lanka's rainfall (1994–1998) visualizes the seasonal shifts that define the country's climatic zones, most notably during the **Southwest Monsoon (May–Sep)**. In this period, the central highlands create a sharp "rain shadow" effect, isolating intense rainfall (>12.6 mm/day) to a southwestern region around Ratnapura while leaving the northern and eastern regions nearly arid. This contrast disappears during the **Second Inter-Monsoon (Oct–Nov)**, the island's wettest overall season, which acts as a critical hydrological reset by bringing widespread heavy rains to the previously dry North and East. The cycle concludes with the **Northeast Monsoon (Dec–Feb)** shifting moisture to the eastern slopes and the transitional **First Inter-Monsoon (Mar–Apr)** returning moderate convection to the southwest, showing that while the Wet Zone enjoys lasting moisture, the Dry Zone is heavily dependent on the year-end monsoonal surge.



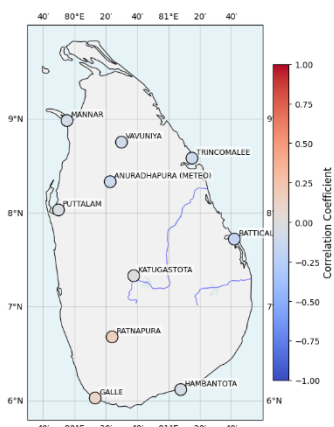
The statistical analysis provided conclusive evidence (with p-values < 0.001) for Sri Lanka's distinct bimodal climate patterns. First, a Kruskal-Wallis test ($\chi^2 = 6124$) confirmed that rainfall varies significantly across the four seasons, rejecting the null hypothesis that seasonal differences are random. Second, the "Rain Shadow" effect was statistically verified by a Mann-Whitney U test, which showed that during the Southwest Monsoon, the Wet Zone (μ : 8.02 mm/day) receives approximately **4.7 times** more rainfall than the Dry Zone (μ : 1.70 mm/day). Finally, the rain deluge phenomenon was validated by a second Mann-Whitney U test, confirming that the Dry Zone undergoes a massive rainfall surge during the Second Inter-monsoon (Oct–Nov), with mean daily rainfall jumping to **8.30 mm**, a figure that statistically exceeds even the Wet Zone's average during the southwest monsoon.

4.4.3 Mechanisms of Variability: The Wind-Rain Correlation Flip

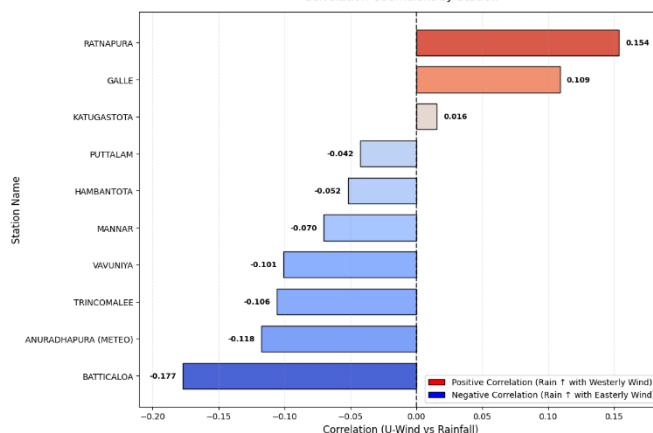
The below visualization reveals a distinct regional split in how zonal wind patterns affect rainfall across Sri Lanka, highlighting a spatial divide driven by the island's topography and seasonal monsoons. Stations in the southwest, such as Ratnapura and Galle, show positive correlations where westerly winds increase precipitation. Conversely, stations in the north and east, notably Batticaloa and Trincomalee, show negative correlations, meaning they receive more rainfall when winds are easterly or when they fall into a dry rain shadow during westerly-dominated periods. While these

patterns align perfectly with known monsoon behavior, the weak magnitude of the coefficients- all staying below 0.2- serves as a reminder that wind direction is just a small factor in the island's complex climate.

Spatial Correlation: U-Wind vs Rainfall in Sri Lanka



Correlation Coefficient by Station



Station	Correlation	P-Value
RATNAPURA	0.153626	1.011010e-20
GALLE	0.108941	4.110918e-11
KATUGASTOTA	0.015558	3.473123e-01
HAMBANTOTA	-0.051842	1.727322e-03
KANKESANTURAI	-0.071668	2.912784e-09
PUTTALAM	-0.074284	6.957494e-15
MULLAITIVU	-0.078884	1.351299e-06
VAVUNIYA	-0.093319	3.205683e-21
TRINCOMALEE	-0.095793	1.236686e-23
JAFFNA	-0.096285	4.467311e-17
MANMAR	-0.100210	1.109568e-22
ANURADHAPURA (METEO)	-0.105061	2.860492e-28
BATTICALOA	-0.176973	1.981347e-26

The statistical analysis confirms that the regional patterns depicted in the charts are mathematically robust and highly significant for nearly all stations. Specifically, the positive correlations in the Southwest (e.g., Ratnapura, $r=+0.154$) and negative correlations in the North/East (e.g., Batticaloa, $r=-0.177$) are statistically verified with extremely low p-values, confirming these are not random occurrences. While Katugastota showed no significant relationship ($p=0.347$), the overall verification demonstrates a consistent, albeit weak, link between zonal wind direction and rainfall distribution

across Sri Lanka.

5. RESULTS AND DISCUSSION

5.1. The Stability of the Historical Rainfall Record:

The analysis of the 30-year rainfall record (1970–2000) reveals that the Dry Zone's hydro-climatology is largely stable, with the Augmented Dickey-Fuller (ADF) confirming that the annual rainfall series is stationary (with a p-value of 0.0014), implying that the statistical properties of rainfall (i.e. mean & variance) have remained largely constant over the period that is studied.

Furthermore, while climate change is an important topic to be addressed in the modern world, this specific dataset did not show any statistically significant monotonic trend in annual rainfall, which was tested by applying a Mann-Kendall test (p-value = 0.165). The validity of this test was also further reinforced by testing for serial correlations in the data using the Ljung-Box test (p-value = 0.297), which confirmed the independence of the annual observations. Finally, a spline regression model was fitted to test for any complex non-linear patterns in the annual rainfall. However, the model p-value was 0.517 which indicates that there is no significant non-linear curvature in the annual totals for rainfall. These findings suggest that rainfall in the Dry Zone is driven by more seasonal short term climate patterns rather than any long-term climate shift.

5.2. Global Drivers:

Analysis of the Niño 3.4 SST variable showed a departure from normality in the SST anomaly distribution which was statistically expressed as well through the application of a Shapiro-Wilk test. Further, a positive skew indicated that there was a stronger push towards 'hotter' El Niño phases during this period. While the El Niño-Southern Oscillation (ENSO) is a known driver of the global climate, our specific statistical analysis of the 1970–2000 period did not find a statistically significant difference in mean rainfall between El Niño & Neutral phases or La Niña & Neutral phases (Mann-Whitney U test). This suggests that the relationship between ENSO and Sri Lankan rainfall may be non-linear or affected by other regional factors.

Further when considering the Southern Oscillation Index, statistically, the Pearson Correlation ($r = 0.038$, $p = 0.48$) and the Independent T-Test ($p = 0.43$) both failed to return significant findings. This coupled with a failure to observe any strong relationships when performing a lag analysis, suggests that SOI has little effect on rainfall in the region.

However, a significant relationship was discovered regarding solar activity. While immediate solar activity showed no difference in the mean amount of rain across solar phases (One-Way ANOVA), a lagged correlation analysis revealed a statistically significant relationship between rainfall and sunspot numbers with a 5-year lag (Spearman rank p-value = 0.0045). This 5-year lag aligns with the half-cycle of the 11-year solar cycle, suggesting that the Dry Zone's rainfall response to solar activity may be delayed, potentially due to the thermal inertia of the oceans or long-term atmospheric feedback loops.

5.3. Regional Drivers: Indian Ocean Dipole (DMI) and All-India Monthly Rainfall

The influence of the Indian Ocean Dipole (DMI) and All-India monthly rainfall on Sri Lankan precipitation was examined using one-sided Mann–Whitney U tests. Rainfall during negative DMI months was compared with positive DMI months, and Sri Lankan rainfall during high India rainfall months was compared with low India rainfall months, separately for dry- and wet-zone stations.

Results showed a significant increase in rainfall during negative DMI months for dry-zone ($p = 0.0066$) and wet-zone ($p = 0.0149$) stations, consistent with the understanding that negative DMI enhances moisture transport toward Sri Lanka. Similarly, high Indian rainfall months were associated with higher Sri Lankan rainfall in both dry- and wet-zone stations, reflecting the influence of regional monsoonal circulation and moisture transport from India. These findings confirm that regional-scale climate factors, including DMI phases and Indian monsoon rainfall, strongly affect Sri Lankan precipitation

5.4. Local Dynamics: The Dominance of the Westerlies

At the local scale, this study looked at the influence of zonal winds (U-winds at 850hPa/1.5km altitude from sea level) on rainfall intensity. The wind data exhibited a clear bimodal distribution, representing the seasonal reversal between the South-West Monsoon (Westerly winds) and the North-East Monsoon (Easterly winds).

Plotting probability of rainfall with U-wind speeds appears to show that extreme rainfall is not greatly influenced by extreme wind speeds but seem to be more influenced by moderate wind speeds. This study also provided statistical confirmation of the dominance of the Westerly winds in bringing rainfall to Sri Lanka. The Mann-Whitney U test yielded a significant result ($p\text{-value} < 0.05$), rejecting the null hypothesis and confirming that Westerly winds (South-West Monsoon) bring significantly higher rainfall magnitudes to the island compared to Easterly winds. This evidence supports the characterization of the South-West Monsoon as one of the primary rainfall drivers for the region.

5.5. Spatio-Temporal Dynamics

The spatio-temporal analysis statistically confirmed Sri Lanka's distinct monsoonal bimodality (Friedman test: $\chi^2 = 93.9395$, $p < 0.001$), which brings drastically different rainfalls to the wet and dry regions. The study verified the "Rain Shadow" effect during the Southwest Monsoon, where the Wet Zone receives approximately 4.7 times more rainfall than the Dry Zone, creating a sharp climatic divide. This contrast dissolves during the Second Inter-monsoon (Oct–Nov), where the Dry Zone experiences a massive hydrological surge that statistically exceeds the Wet Zone's monsoon averages. Finally, a "Wind-Rain Correlation Flip" was observed: westerly winds correlate positively with rainfall in the southwest ($r = +0.154$) but negatively in the east ($r = -0.177$), showing that wind influence is geographically non-uniform.

6. CONCLUSION

This study addressed the "climatological isolation" of Sri Lanka's historical rainfall records by analyzing augmented datasets from different subset time regions in 1970 to 2000. The analysis of the 30-year record revealed a stable hydro-climatology with no significant monotonic trends or non-linear curvature, suggesting that Dry Zone variability is driven by short-term seasonal cycles rather than long-term climate patterns.

In terms of drivers, global factors such as ENSO and SOI showed no immediate statistical impact on rainfall, though solar activity demonstrated a significant 5-year lagged relationship. Conversely, regional drivers proved robust; negative Indian Ocean Dipole phases and high Indian rainfall were significantly associated with increased precipitation in Sri Lanka.

Locally, the Southwest Monsoon (Westerly winds) was confirmed as a primary rainfall driver. However, this influence is spatially distinct: the "Rain Shadow" restricts moisture to the Wet Zone mid-year, leaving the Dry Zone dependent on the rainfall surge of the Second Inter-monsoon. Ultimately, the study concludes that Sri Lankan rainfall is defined less by global planetary indices and more by the complex interaction of regional monsoonal dynamics and local topography.

7. REFERENCES

1. Nino3.4 SST from the Hadley Centre Sea Ice and Sea Surface Temperature dataset(<https://www.metoffice.gov.uk/hadobs/hadisst/>)
2. Southern Oscillation Index (SOI) 3 decimal place dataset made available by the Climatic Research Unit, University of East Anglia(<https://crudata.uea.ac.uk/cru/data/soi/>)
3. WDC-SILSO dataset, Royal Observatory of Belgium(<https://sidc.be/SILSO/datafiles>)
4. Dipole Mode Index (DMI) Monthly Time-series dataset - National Oceanic and Atmospheric Administration derived from the HadISST1.1 Dataset (<https://psl.noaa.gov/data/timeseries/month/data/dmi.had.long.data>)
5. All-India and Macro-Regional dataset- Indian Institute of Tropical Meteorology (<https://www.tropmet.res.in/data/data-archival/rain/iitm-regionrf.txt>)
6. NOAA-CIRES-DOE 20th Century Reanalysis V3 dataset (https://www.psl.noaa.gov/data/gridded/data.20thC_ReanV3.html)
7. What is El Niño-Southern Oscillation (ENSO)? *National Oceanic & Atmospheric Administration*(<https://www.climate.gov/news-features/blogs/enso/what-el-nino-southern-oscillation-enso-nutshell>)
8. Indian Ocean Dipole (IOD) - Relatively New Natural Phenomenon Identified in Indian Ocean which is Influencing Various Ways to Sri Lanka: *Institute of National Security Studies*(<https://www.inss.lk/index.php?id=671>)

ACKNOWLEDGEMENT

We would like to sincerely thank Dr. H. K. W. I. Jayawardena (Senior Lecturer-Department of Physics, Open University of Sri Lanka) for providing our group with the important 100-year historical Sri Lanka rainfall data that was used to build up the dataset which was used in this study, as well as all the specific domain knowledge that she provided, which enabled us to complete this study in a more robust manner.